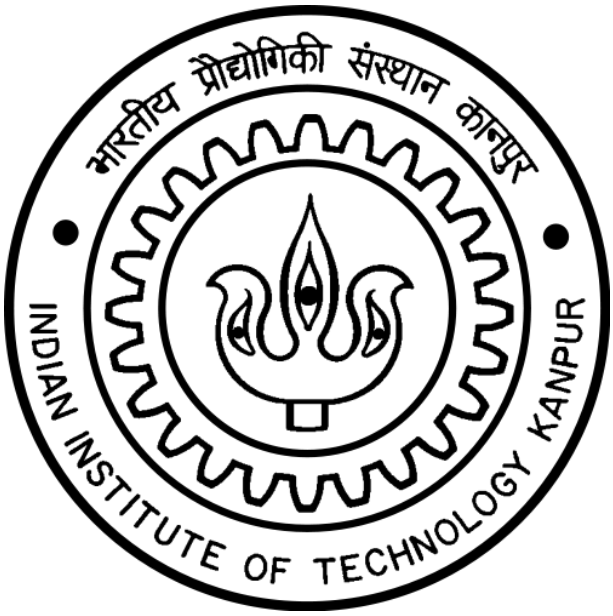


SPA617

RADIO ASTRONOMY LABORATORY

RF Transmission, Reflections, Standing Waves

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1 RF Transmission, Reflections, and Standing Waves

1.1 Aim

To understand signal propagation, reflection, and standing-wave formation in a coaxial transmission line at radio frequencies, and to determine important cable parameters experimentally.

1.2 Objectives

(a) To determine the phase velocity and velocity factor of a signal in a coaxial cable from the frequency spacing between consecutive maxima or minima produced by reflection from an open-ended cable.

(b) To study signal reflection from an open-ended cable and from a cable terminated with a $50\,\Omega$ load.

(c) To estimate the magnitude of the reflection coefficient and verify that the coaxial cable is approximately matched to a $50\,\Omega$ termination, indicating that its characteristic impedance is close to $50\,\Omega$.

(d) To calculate the inductance and capacitance per unit length using the measured phase velocity and the approximate value $Z_0 \approx 50\,\Omega$.

(e) To measure the insertion loss and attenuation per unit length of the coaxial cable.

1.3 Theory

1.3.1 Transmission lines and reflections

A transmission line is a structure that guides electromagnetic energy from one point to another. A coaxial cable supports transverse electromagnetic (TEM) propagation, for which both the electric and magnetic fields are transverse to the direction of propagation. The signal velocity in a cable is lower than the speed of light in vacuum because of the dielectric material between the conductors.

When a travelling wave encounters a load impedance Z_L that is different from the characteristic impedance Z_0 of the cable, a part of the wave is reflected toward the source. At any fixed observation point, the incident and reflected waves superimpose. Depending on their relative phase, the measured amplitude may be large because of constructive interference or small because of destructive interference.

In this experiment, the oscilloscope measures the voltage at one fixed point, namely the Tee junction. It does not directly display the spatial standing-wave pattern along the cable. When the signal frequency is varied, the round-trip phase of the reflected wave changes, causing the measured amplitude at the Tee junction to vary periodically.

1.3.2 Phase velocity measurement

Let the physical length of the coaxial cable be l and the phase velocity in the cable be v_p. The reflected signal travels a round-trip distance $2l$, so the round-trip time delay is

$$\tau = \frac{2l}{v_p} \tag{1}$$

The round-trip phase of the reflected signal is

$$\phi_{\mathrm{RT}} = 2\pi f\tau + \phi_\Gamma = \frac{4\pi f l}{v_p} + \phi_\Gamma, \tag{2}$$

where ϕ_Γ is the phase shift introduced by the termination. For an ideal open circuit, $\phi_\Gamma = 0$.

Two consecutive maxima, or two consecutive minima, are separated by a round-trip phase change of 2π. Therefore,

$$2\pi\Delta f\tau = 2\pi, \tag{3}$$

which gives

$$\tau = \frac{1}{\Delta f} \tag{4}$$

and

$$v_p = 2l\,\Delta f \tag{5}$$

The velocity factor of the cable is

$$\mathrm{VF} = \frac{v_p}{c} \tag{6}$$

where $c \approx 3 \times 10^8\,\mathrm{m/s}$ is the speed of light in vacuum.

Here, Δf is the frequency difference between two consecutive maxima or two consecutive minima.

1.3.3 Reflection coefficient and characteristic impedance

The voltage reflection coefficient at the load is

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} \tag{7}$$

where Z_L is the load impedance and Z_0 is the characteristic impedance of the cable.

For important special cases:

- Open circuit: $Z_L \to \infty$ and $\Gamma = +1$.
- Short circuit: $Z_L = 0$ and $\Gamma = -1$.
- Matched load: $Z_L = Z_0$ and $\Gamma = 0$.

During a frequency sweep, the oscilloscope displays the superposition of the incident and reflected waves:

$$V_{\mathrm{osc}} = V_{\mathrm{incident}} + V_{\mathrm{reflected}}. \tag{8}$$

When the incident and reflected waves are in phase, they interfere constructively and the measured amplitude is maximum:

$$V_{\mathrm{max}} = |V_{\mathrm{incident}}| + |V_{\mathrm{reflected}}|. \tag{9}$$

When the incident and reflected waves are out of phase by 180°, they interfere destructively and the measured amplitude is minimum. Assuming the incident-wave amplitude is greater than or equal to the reflected-wave amplitude,

$$V_{\mathrm{min}} = |V_{\mathrm{incident}}| - |V_{\mathrm{reflected}}|. \tag{10}$$

Adding and subtracting these two equations gives

$$|V_{\mathrm{incident}}| = \frac{V_{\mathrm{max}} + V_{\mathrm{min}}}{2} \tag{11}$$

and

$$|V_{\mathrm{reflected}}| = \frac{V_{\mathrm{max}} - V_{\mathrm{min}}}{2}. \tag{12}$$

Therefore, the magnitude of the reflection coefficient observed at the Tee junction can be estimated as

$$|\Gamma| = \frac{|V_{\mathrm{reflected}}|}{|V_{\mathrm{incident}}|} = \frac{V_{\mathrm{max}} - V_{\mathrm{min}}}{V_{\mathrm{max}} + V_{\mathrm{min}}}. \tag{13}$$

In this experiment, only a $50\,\Omega$ termination is used. The reflected signal obtained with this termination is compared with that obtained when the cable is left open.

1.3.4 Inductance and capacitance per unit length

Let L' denote the inductance per unit length in H/m and C' denote the capacitance per unit length in F/m. For a low-loss transmission line,

$$v_p = \frac{1}{\sqrt{L'C'}} \tag{14}$$

and

$$Z_0 = \sqrt{\frac{L'}{C'}}. \tag{15}$$

Therefore,

$$L' = \frac{Z_0}{v_p} \tag{16}$$

and

$$C' = \frac{1}{Z_0 v_p}. \tag{17}$$

1.3.5 Cable attenuation

Cable attenuation is the reduction in signal level as the wave propagates through the cable. In this experiment, power is measured in dBm using a spectrum analyser. The insertion loss of the cable is

$$\mathrm{IL} = P_{\mathrm{ref}} - P_{\mathrm{cable}}, \tag{18}$$

where P_{ref} is the power measured using a short reference connection and P_{cable} is the power measured after inserting the cable under test. The attenuation per unit length is then

$$\alpha = \frac{\mathrm{IL}}{l}\ \mathrm{dB/m}. \tag{19}$$

1.4 Components Required

- RF signal generator
- Digital storage oscilloscope (DSO)
- Spectrum analyser
- RF Tee connector
- Coaxial cable under test
- Short reference coaxial cable
- $50\,\Omega$ termination
- Open-circuit termination, obtained by leaving the cable end unconnected
- Suitable RF adapters and connectors

1.5 Experimental Setup

Connect the signal generator, cable under test, and oscilloscope through the Tee connector as shown in Fig. 1. The far end of the cable is terminated according to the measurement being performed.

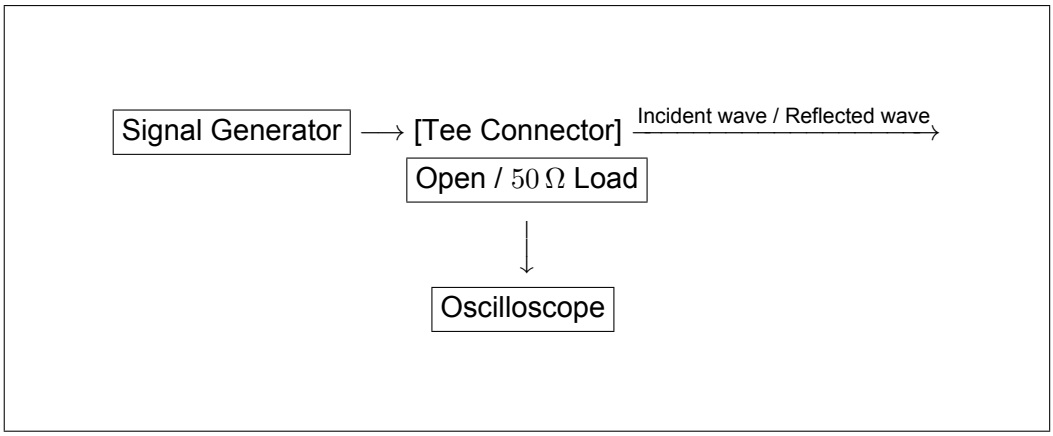


Figure 1: Experiment Setup.

1.6 Experimental Procedures

1.6.1 Part A: Phase velocity and velocity factor

1. Measure and record the physical cable length l.
2. Connect the signal generator, Tee connector, cable, and oscilloscope as shown in Fig. 1.
3. Leave the far end of the cable open.
4. Set the oscilloscope input impedance to $1\,\mathrm{M}\Omega$.
5. Apply a sinusoidal signal of constant amplitude.
6. Sweep the frequency slowly over a range containing at least two maxima and two minima. Record the signal amplitude at suitable frequency intervals.
7. Plot the measured signal amplitude versus frequency and identify the frequencies f_1 and f_2 of two consecutive maxima or two consecutive minima.
8. Calculate the frequency spacing:

$$\Delta f = |f_2 - f_1| \tag{20}$$

   Convert Δf to hertz before using it in the phase-velocity calculation.

9. Calculate the phase velocity and velocity factor:

$$v_p = 2l\,\Delta f, \qquad \mathrm{VF} = \frac{v_p}{c} \tag{21}$$

1.6.2 Part B: Reflection coefficient and approximate characteristic impedance

1. Retain the Tee-junction setup and use the same frequency range employed for the open-circuit measurement in Part A.
2. From the open-circuit data, determine $V_{\mathrm{max,open}}$ and $V_{\mathrm{min,open}}$ over one complete ripple period.
3. Calculate the magnitude of the reflection coefficient for the open-ended cable:

$$|\Gamma_{\mathrm{open}}| = \frac{V_{\mathrm{max,open}} - V_{\mathrm{min,open}}}{V_{\mathrm{max,open}} + V_{\mathrm{min,open}}}. \tag{22}$$

4. Connect the $50\,\Omega$ termination to the far end of the cable.
5. Sweep over the same frequency range without changing the signal-generator amplitude.
6. Record $V_{\mathrm{max},50}$ and $V_{\mathrm{min},50}$.
7. Calculate
$$|\Gamma_{50}| = \frac{V_{\mathrm{max},50} - V_{\mathrm{min},50}}{V_{\mathrm{max},50} + V_{\mathrm{min},50}}. \tag{23}$$
8. Compare $|\Gamma_{\mathrm{open}}|$ and $|\Gamma_{50}|$.

Table 1: Reflection coefficient measurements for different terminations.

Termination	V_{max} (V)	V_{min} (V)	$\lvert\Gamma\rvert$
Open circuit	1.72	0.125	0.864
$50\,\Omega$ load	0.928	0.900	0.015

1.6.3 Part C: Inductance and capacitance per unit length

$L' = 24.43\ \mu$H/m and $C' = 103.89$ pF/m.

1.6.4 Part D: Cable attenuation

1. Connect the signal generator to the spectrum analyser using a short reference cable and the required adapters.
2. Set a suitable test frequency and input power. Keep both fixed throughout the measurement.
3. Record the reference power P_{ref} in dBm.
4. Replace the short reference connection with the cable under test while keeping the remaining setup unchanged.
5. Record the measured power P_{cable} in dBm.
6. Calculate the cable insertion loss:
$$\mathrm{IL} = P_{\mathrm{ref}} - P_{\mathrm{cable}}. \tag{24}$$
7. Calculate the attenuation per unit length:
$$\alpha = \frac{\mathrm{IL}}{l}\ \mathrm{dB/m}. \tag{25}$$
8. Repeat at additional frequencies to study the frequency dependence of attenuation.

Table 2: Power measurements and attenuation calculation.

Frequency (MHz)	P_{ref} (dBm)	P_{cable} (dBm)	α (dB/m)
100	-0.57	-0.60	0.032
150	-0.79	-0.72	0.076
200	-0.91	-0.90	0.011

Table 3: Summary of experimental results and calculated cable parameters.

Quantity	Symbol	Measured/Calculated Value
Cable length (m)	l	0.925
Frequency spacing (MHz)	Δf	116
Phase velocity (m/s)	v_p	2.148 x 10^8
Velocity factor	v_p/c	0.716
Approximate characteristic impedance (Ω)	Z_0	48.5
Inductance per unit length (μH/m)	L'	24.43
Capacitance per unit length (pF/m)	C'	103.89
Attenuation per unit length (dB/m, 150Hz)	α	0.076

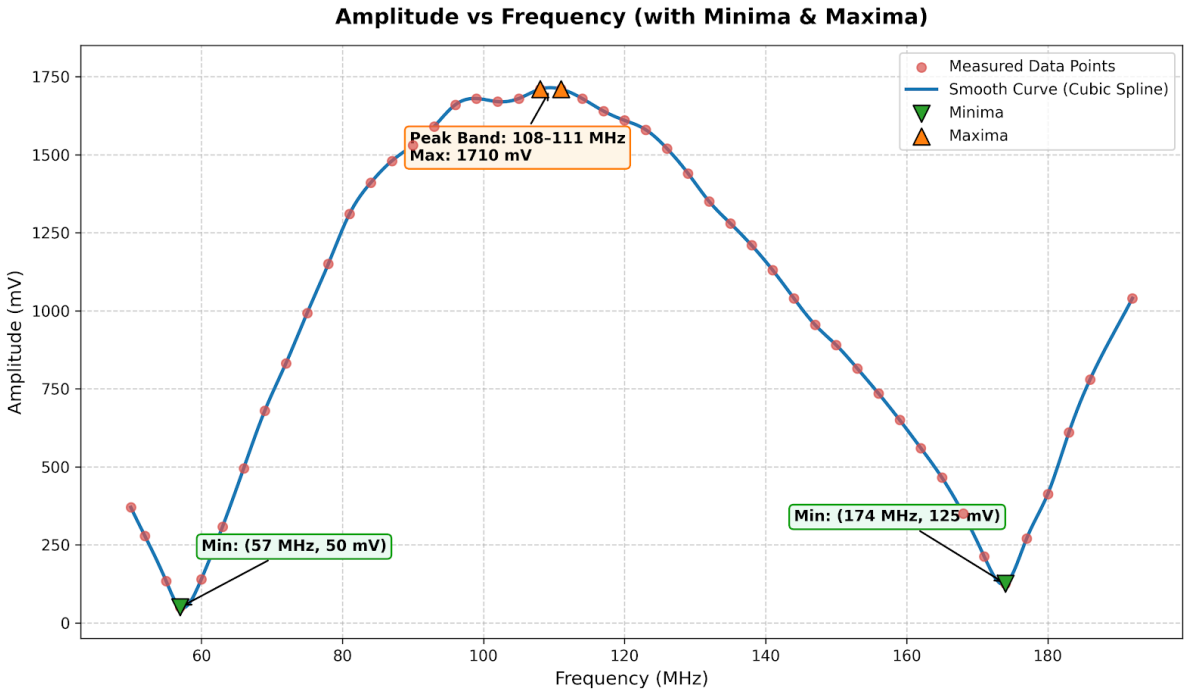


Figure 2: Variation of standing-wave signal amplitude with frequency

1.7 Conclusions

1. From the plot showing the variation of standing-wave signal amplitude with frequency due to interference, we observe that the signal amplitude measured fluctuates periodically between maxima and minima as the frequency is swept. The oscilloscope observes the superposition of the incident wave and the wave reflected from the far end of the cable ($V_{\text{osc}} = V_{\text{incident}} + V_{\text{reflected}}$). Because the cable has a fixed physical length l, varying the frequency continuously shifts the round-trip electrical phase $\phi_{\text{RT}} = \frac{4\pi f l}{v_p} + \phi_\Gamma$. Constructive interference occurs whenever the round-trip phase shift is an integer multiple of 2π, producing amplitude peaks, while destructive interference produces periodic nulls/troughs.

2. The phase velocity v_p represents the speed at which electromagnetic wave phase fronts propagate through the cable. Because the coaxial line is filled with a dielectric (typically solid or foamed polyethylene, $\varepsilon_r \approx 2.25$), $v_p = \frac{c}{\sqrt{\varepsilon_r}}$ is strictly less than the speed of light in vacuum ($c \approx 3 \times 10^8$ m/s).

3. For the open circuit ($Z_L \to \infty$), total reflection occurs ($\Gamma \to +1$), producing large peak-to-peak ripple between V_{max} and V_{min} ($|\Gamma_{\text{open}}| \approx 1$). When terminating the line with the $50\ \Omega$ load, the ripple amplitude collapses substantially ($V_{\text{max}} \approx V_{\text{min}}$), driving $|\Gamma_{50}| \approx 0$. This substantial reduction in reflected power confirms that $Z_L \approx Z_0$, verifying that the characteristic impedance of the transmission line is approximately $50\ \Omega$.

4. Attenuation quantifies the exponential signal decay per meter due to physical loss mechanisms.

   - **Conductor losses (ohmic/skin effect)**: Proportional to $\sqrt{f}$ due to current concentration along the outer skin depth of the inner and outer conductors.
   - **Dielectric losses**: Proportional to $f \cdot \tan\delta$, arising from alternating polarization dissipation in the insulating medium

5. Possible sources of measurement uncertainty are -

   - **Oscilloscope Loading and Input Capacitance** : Although the DSO input resistance is set to $1\ \text{M}\Omega$, its parallel input capacitance ($\approx$ 10–20 pF) shunts high-frequency signals at the Tee junction, slightly loading the line.
   - **Connector and Discontinuity parasitics** : The RF Tee adapter, BNC/SMA barrel connectors, and open-cable fringing field introduce stray reactances and minor phase offsets ($\phi_\Gamma \neq 0$).
   - **Finite Cable Length Measurement (l)** : Inaccuracies in determining the exact electrical length of the cable (including internal pin depths of end connectors) propagate linearly into v_p, L', C', and α.
   - **Frequency Resolution and Peak Identification** : Uncertainty in pinpointing the exact crests and troughs (f_1, f_2) on the DSO amplitude sweep directly impacts Δf.
   - **Cable Loss Impact on $|\Gamma_{\text{open}}|$**: Cable attenuation reduces the round-trip reflected amplitude reaching the Tee junction, causing $|\Gamma_{\text{open}}|$ to evaluate to slightly less than the ideal theoretical value of 1.